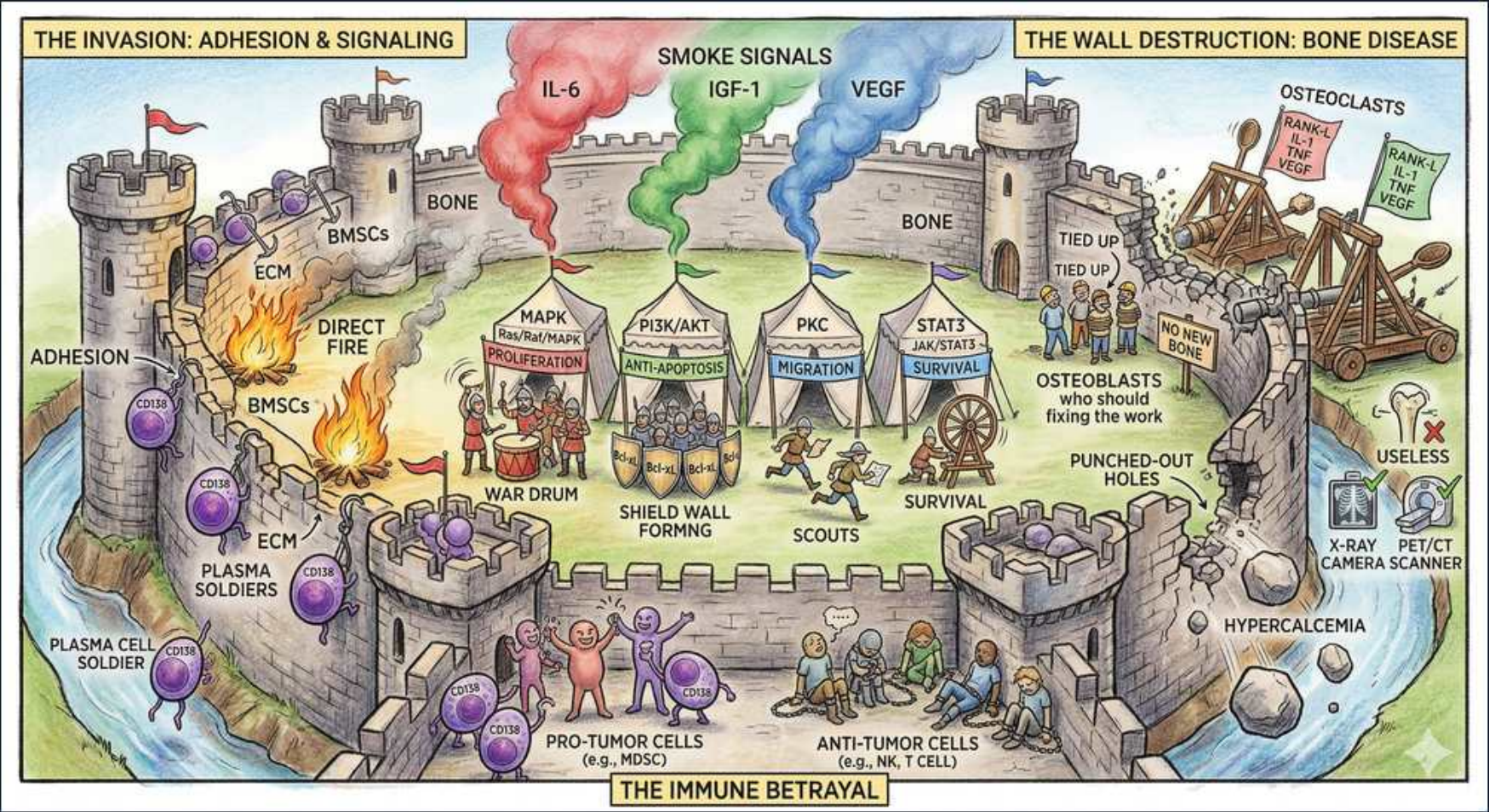


Myeloma Bone Disease & Signaling — The Castle Siege



1. Plasma cell adhesion to BMSCs

WHAT YOU SEE

Plasma-cell soldiers gripping the castle wall via ECM.

WHAT IT ENCODES

Malignant plasma cells adhere to bone marrow stromal cells (BMSCs) through adhesion molecules and the extracellular matrix. This contact triggers cytokine release and survival signals.

BOARD TRAP

Adhesion to the stroma (not autonomous growth) drives myeloma — the marrow microenvironment is essential, which is why the niche is a therapeutic target.

2. IL-6 (key survival cytokine)

WHAT YOU SEE

Red 'IL-6' smoke signal rising.

WHAT IT ENCODES

BMSCs secrete IL-6, the central growth/survival cytokine for myeloma cells, signaling through JAK/STAT3 to promote proliferation and resist apoptosis.

BOARD TRAP

IL-6 is the dominant myeloma survival cytokine (via STAT3) — distinguish it from VEGF (angiogenesis) and IGF-1.

3. IGF-1 signaling

WHAT YOU SEE

Green 'IGF-1' smoke signal.

WHAT IT ENCODES

IGF-1 activates PI3K/AKT, promoting myeloma cell migration and anti-apoptotic survival, reinforcing the IL-6 growth program.

BOARD TRAP

IGF-1 → PI3K/AKT drives survival/migration; pairing the cytokine with its pathway is a favorite exam linkage.

4. VEGF (angiogenesis)

WHAT YOU SEE

Blue 'VEGF' smoke signal.

WHAT IT ENCODES

VEGF promotes marrow angiogenesis, supplying the expanding plasma-cell clone, and feeds back to support tumor growth.

BOARD TRAP

In myeloma VEGF means neovascularization; don't confuse with POEMS where VEGF is the central diagnostic biomarker.

5. MAPK (Ras/Raf/MEK) — proliferation

WHAT YOU SEE

War drum: MAPK Ras/Raf/MEK 'proliferation.'

WHAT IT ENCODES

Cytokine signaling activates the MAPK (Ras-Raf-MEK-ERK) cascade driving proliferation. RAS mutations are common and contribute to disease progression.

BOARD TRAP

MAPK = proliferation; STAT = survival; PI3K/AKT = survival/migration — match each pathway to its output.

6. PI3K/AKT — anti-apoptosis/migration

WHAT YOU SEE

Scouts marker: PI3K/AKT 'migration.'

WHAT IT ENCODES

PI3K/AKT signaling promotes anti-apoptosis and cell migration/homing, helping plasma cells survive and spread within the marrow.

BOARD TRAP

PI3K/AKT supports survival AND migration — a stress/death pathway misattribution is the trap here.

7. STAT3 — survival

WHAT YOU SEE

Banner: STAT3/STAT1 'survival.'

WHAT IT ENCODES

IL-6 activates JAK/STAT3, upregulating anti-apoptotic proteins (e.g., Mcl-1, Bcl-xL) for plasma-cell survival — the core IL-6 effect.

BOARD TRAP

STAT3 is the IL-6 downstream survival node; blocking IL-6/STAT3 removes the key anti-apoptotic signal.

8. Osteoclast activation (RANKL)

WHAT YOU SEE

Catapults ('RANK the wall') attacking bone.

WHAT IT ENCODES

Myeloma shifts the RANKL:OPG ratio toward RANKL, activating osteoclasts that resorb bone. This produces the lytic lesions and releases calcium.

BOARD TRAP

Increased RANKL/decreased OPG drives the lytic disease — denosumab (anti-RANKL) and bisphosphonates target this axis.

9. Osteoblast suppression (DKK1)

WHAT YOU SEE

Osteoblasts 'tied up,' not fixing the wall.

WHAT IT ENCODES

Myeloma cells secrete DKK1 and other inhibitors that SUPPRESS osteoblasts via the Wnt pathway, so bone is not rebuilt — explaining why lesions are purely lytic and don't heal.

BOARD TRAP

Both arms are hit: osteoclasts up AND osteoblasts suppressed (DKK1/Wnt) — that's why myeloma bone scans are 'cold' and lesions don't sclerose/heal.

10. Punched-out lytic lesions

WHAT YOU SEE

Holes punched in the castle wall.

WHAT IT ENCODES

The net effect is classic 'punched-out' lytic bone lesions on skeletal survey, with pathologic fractures and bone pain.

BOARD TRAP

Lytic ('punched-out') lesions are the myeloma signature — opposite of the sclerotic lesions in POEMS.

11. Bone scan is falsely negative; use X-ray/PET-CT

WHAT YOU SEE

X-ray / PET-CT scanners; 'useless' nuclear bone scan.

WHAT IT ENCODES

Because osteoblasts are suppressed, technetium bone scans are insensitive (cold) in myeloma. Imaging uses skeletal survey, low-dose whole-body CT, MRI, or PET-CT.

BOARD TRAP

Standard radionuclide bone scan MISSES myeloma lesions — order skeletal survey/CT/MRI/PET-CT, not a Tc bone scan.

12. Hypercalcemia

WHAT YOU SEE

Hypercalcemia flag near the broken wall.

WHAT IT ENCODES

Osteoclastic bone resorption releases calcium, causing hypercalcemia (the 'C' of CRAB) with confusion, constipation, polyuria, and renal injury.

BOARD TRAP

Myeloma hypercalcemia is resorptive (bone breakdown), so PTH is suppressed — distinguish from primary hyperparathyroidism.

13. Pro-tumor cells (MDSCs)

WHAT YOU SEE

Pro-tumor figures (e.g., MDSCs) inside the castle.

WHAT IT ENCODES

Myeloid-derived suppressor cells and other pro-tumor populations create an immunosuppressive microenvironment that protects the clone from immune clearance.

BOARD TRAP

The marrow niche is immunosuppressive (MDSCs) — 'the immune betrayal,' explaining immune evasion and rationale for immunotherapy.

14. Anti-tumor cells suppressed (NK/T cells)

WHAT YOU SEE

Anti-tumor figures (NK, T cells) outnumbered.

WHAT IT ENCODES

Effective NK- and T-cell responses are blunted in the myeloma microenvironment, allowing tumor escape — the basis for CAR-T and bispecific (BCMA) therapies.

BOARD TRAP

Failing NK/T-cell immunity (not just intrinsic growth) lets myeloma persist — targeted by BCMA CAR-T/bispecifics.

15. Direct anti-apoptotic 'direct fire'

WHAT YOU SEE

Direct fire / war drum at the wall.

WHAT IT ENCODES

Beyond paracrine signals, adhesion delivers direct anti-apoptotic and proliferative signals to plasma cells (cell-adhesion-mediated drug resistance), reducing chemosensitivity.

BOARD TRAP

Stromal adhesion confers drug resistance (CAM-DR) — a reason single agents fail and combination/niche-disrupting therapy is used.
